



Guseyn Ismayylov

Senior Full-Stack Software Engineer · Technical Leader

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Senior Full-Stack AI Software Engineer (Node.js) with **10 years** building production platforms end to end — Node.js backends, React/Next.js frontends, PostgreSQL/Redis data layers, and AWS cloud infra. Deep **Node.js** expertise (Express, Fastify, NestJS); author of nodes.js backend framework. Production **AI** work: agentic backends, LLM middleware, RAG pipelines, n8n automations, and AI-native delivery via **Cursor** (multi-model Composer) and **Google Gemini**. Also built EHTML and Unison. Applied Mathematics background; author of Mathematics of Neural Networks.

SKILLS

Node.js (Primary) — Express, Fastify, NestJS, TypeScript/JavaScript, async/event-loop, streams, worker threads & cluster, HTTP/2, REST/GraphQL APIs, microservices, BFF patterns, serverless (AWS Lambda), custom framework design (nodes.js).

AI & Intelligent Backends — LLM middleware & agent orchestration on Node.js, RAG pipelines, vector search & semantic retrieval, MCP/tool-calling APIs, streaming inference, prompt/tool-chain design, guardrails & eval for production AI, n8n workflow automation, event-driven AI microservices.

AI-Native Development — Cursor IDE (Composer — Claude Opus/Sonnet, GPT, Gemini, Grok model routing), Google Gemini (API & workspace) — AI-accelerated Node.js/TypeScript delivery, autonomous refactors, and test generation.

Generative AI Stack — OpenAI, Anthropic Claude, Google Gemini APIs; embeddings, function calling, structured outputs; Node.js LLM middleware patterns; Redis/DynamoDB/S3 hybrid retrieval tiers; token & latency optimization.

System Architecture — Distributed Systems, Microservices, Event-Driven Architecture (EDA), Multi-tenant SaaS, High-Load MVPs, System Design, Scalability, High Availability & Reliability Engineering.

Backend Ecosystem — Node.js (Express, Fastify, NestJS), JavaScript/TypeScript, Java (Spring Boot), PHP, Python (FastAPI), NX Monorepos, RESTful APIs, GraphQL, Serverless Architecture.

High-Performance UI/UX — Web Components, EHTML (Creator), React, Next.js, React Server Components (RSC), React Context, React Query, Tailwind CSS, Material UI, Zustand, Angular, Real-time SVG/Canvas Rendering, Shadow DOM, Micro-frontends, HTML5/CSS3, Vite, Esbuild.

Data Engineering & Storage — PostgreSQL, MySQL, MariaDB, Redis (Caching, Pub/Sub), ClickHouse (OLAP), Snowflake, Supabase, Drizzle ORM, DynamoDB, Prisma, Apache Kafka, RabbitMQ, MongoDB,

ElasticSearch, Database Optimization.

Cloud Platforms & DevOps (IaC) — AWS (EC2, ECS, S3, RDS, Lambda, DynamoDB, CloudWatch), GCP (Cloud Run, App Engine, BigQuery, VPC), Azure, DigitalOcean, Docker, Kubernetes (K8s), Terraform, GitHub Actions, GitLab CI/CD, Infrastructure as Code (IaC).

Observability & Telemetry — OpenTelemetry, Prometheus, Grafana, New Relic, Graylog, Datadog, AWS CloudWatch, ELK/Elastic Stack.

Protocols & Integrations — WebRTC (Xirsys), WebSockets, Stripe API (Payment Infrastructure), Postmark (Email Delivery), Azure AD (OAuth2/OIDC), Secure Identity Management.

Testing & Quality Assurance — Jest, Vitest, Supertest, Testcontainers, Playwright, LLM eval suites & prompt regression, React Testing Library, API contract tests, Postman/Newman, TDD/BDD, coverage & reporting (Allure, Istanbul), CI test gates, parallel test runs, load/performance testing.

AI — Cursor (Composer — Claude, GPT, Gemini, Grok models), Google Gemini, n8n; LLM-assisted coding, review, and refactoring.

WORK EXPERIENCE

InstruxMusic — Senior Full-Stack Developer, Tech Lead

Nov 2024 – Present · Remote · USA

- Architected and accelerated the launch of a complex ed-tech ecosystem, delivering a production-ready MVP in under 3 months using a streamlined stack of PostgreSQL, Docker, and DigitalOcean.
- Engineered a scalable multi-tenant scheduling engine and automated invoicing system, integrating Stripe Payments to establish seamless transactional workflows and eliminate manual billing overhead.
- Developed a proprietary, low-latency video conferencing solution from scratch using vanilla Web Components and Xirsys (STUN/TURN), bypassing heavy third-party framework dependencies to achieve highly efficient real-time streaming, chat, and data channel communication.
- Designed high-deliverability notification pipelines by implementing Postmark, significantly cutting down user communication friction through automated transactional workflows.
- Drive technical strategy and product vision in alignment with business scaling and pricing frameworks, managing execution speed while maintaining a high-performance, low-bloat architecture.

[View Recommendations](#)

Congrify — Full Stack Senior Software Engineer, Tech Lead

Oct 2023 – Oct 2024 · Remote · Germany/Italy

- Directed technical strategy and system observability by architecting a specialized monitoring system using Grafana and AWS CloudWatch, enabling real-time detection of payment transaction anomalies and resolving critical infrastructure bottlenecks.

- Reduced AWS Lambda operational costs by 20% through targeted workflow optimizations, memory tuning, and improved data-processing logic across serverless environments.
- Spearheaded the modernization of legacy microservices, dramatically improving concurrency patterns, query execution, and high-load stability by migrating complex datasets to ClickHouse (OLAP) and Snowflake.
- Enhanced large-scale data ingestion and asynchronous messaging pipelines by leveraging AWS SQS and robust Pub/Sub architectures, ensuring zero-loss system reliability and high availability under heavy transactional surges.
- Engineered a robust AI middleware layer alongside low-latency data tiers utilizing DynamoDB and S3 to facilitate real-time streaming data processing and intelligent user analytical insights.
- Led an end-to-end frontend redesign and streamlined communication systems by integrating AWS SES for transactional mailings and optimizing the React ecosystem with advanced caching and sophisticated state management.

FxPro — Senior Software Engineer

Nov 2022 – Oct 2023 · Limassol, Cyprus

- Orchestrated infrastructure for 60+ microservices, delivering a 20% improvement in system reliability and a 15% reduction in operational downtime.
- Created a centralized Authentication Service with Azure Active Directory integration and successfully migrated payment configurations for 20+ global systems across 50+ countries.
- Optimized data consistency and performance across distributed services by leveraging Redis for caching and PostgreSQL for mission-critical financial data.
- Streamlined CI/CD pipelines using Artifactory in GitLab, reducing deployment cycles by 25% and ensuring high-availability across global environments.
- Implemented full-stack observability across 25+ services using New Relic, Graylog, and Grafana, enabling real-time bottleneck identification and rapid incident response.

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Cognitran — Senior Software Engineer

Mar 2021 – Oct 2022 · Krakow, Poland

Developed mission-critical, offline-first automotive diagnostic software and automated technical documentation pipelines for enterprise clients including Ford and Jaguar.

Huawei — Senior Software Engineer

Jan 2020 – Mar 2021 · Moscow, Russia

Designed high-performance cloud infrastructure components, optimized internal developer tooling, and crafted high-throughput custom data parsers.

US Mobile — Senior Software Engineer

Feb 2019 – Jan 2020 · Remote · USA

- Helped design technical solutions and created a roadmap for migrating infrastructure to Google Cloud Platform (GCP).

- Successfully oversaw the implementation of this replatforming, ensuring a smooth and efficient transition.
- Spearheaded the full automation of report processing from third-party sources, utilizing GCP services to significantly improve data processing and operational efficiency.

Grid Dynamics — Intern to Senior Software Engineer

Dec 2015 – Jan 2019

Progressed from Intern to Senior Engineer, delivering high-impact cloud-native R&D internal tooling for major enterprise clients including Google, Raymond James, and Home Depot.

CERTIFICATIONS

Oracle Certified Associate Java SE 8 Programmer

OPEN SOURCE

EHTML, Nodes.js, e-ui, Mathematics of Neural Networks

EDUCATION

Applied Mathematics and Computer Science — *Saratov State University, 2012 - 2016*

Bachelor's Thesis (PDF).